

A



B

A

B

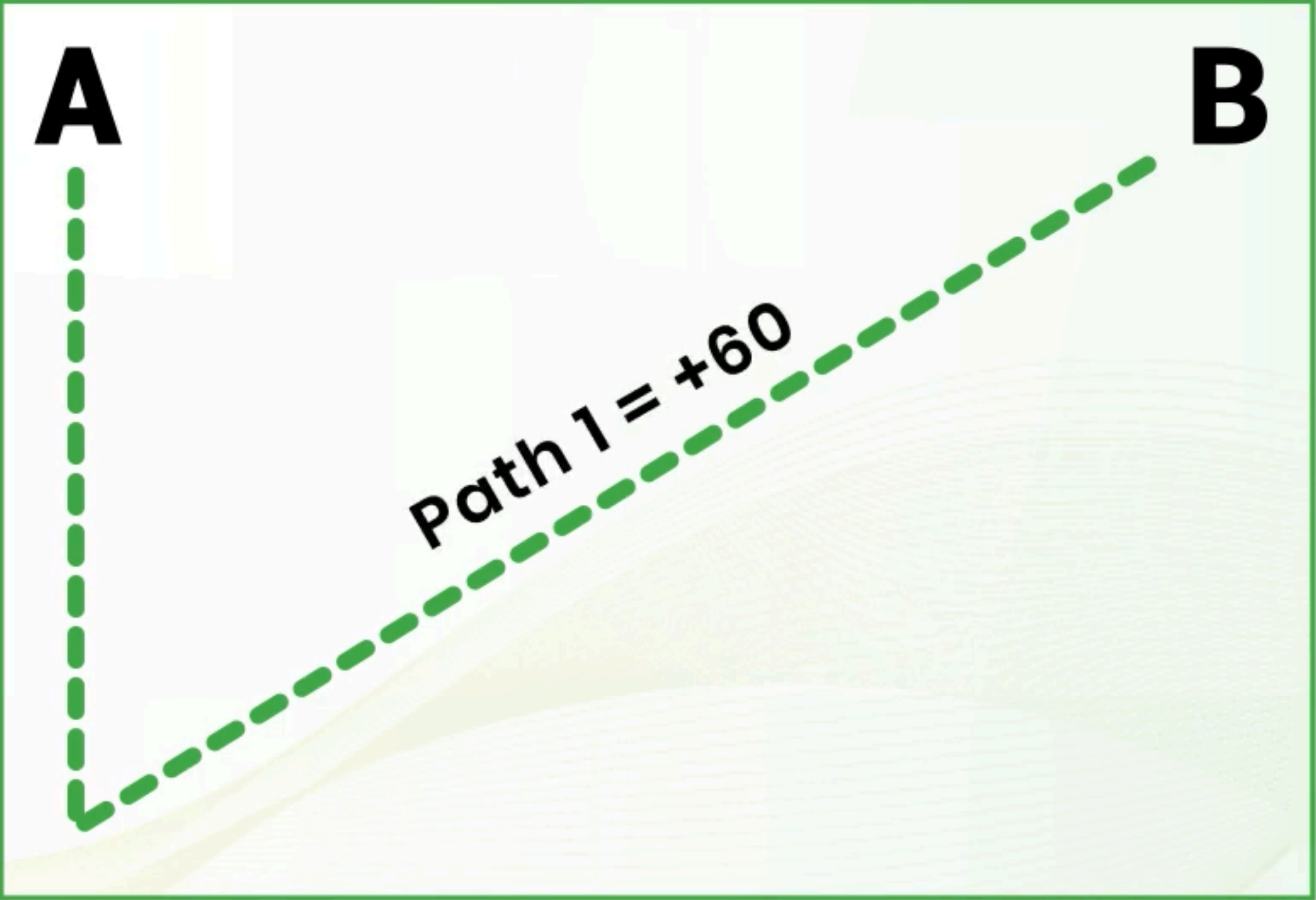


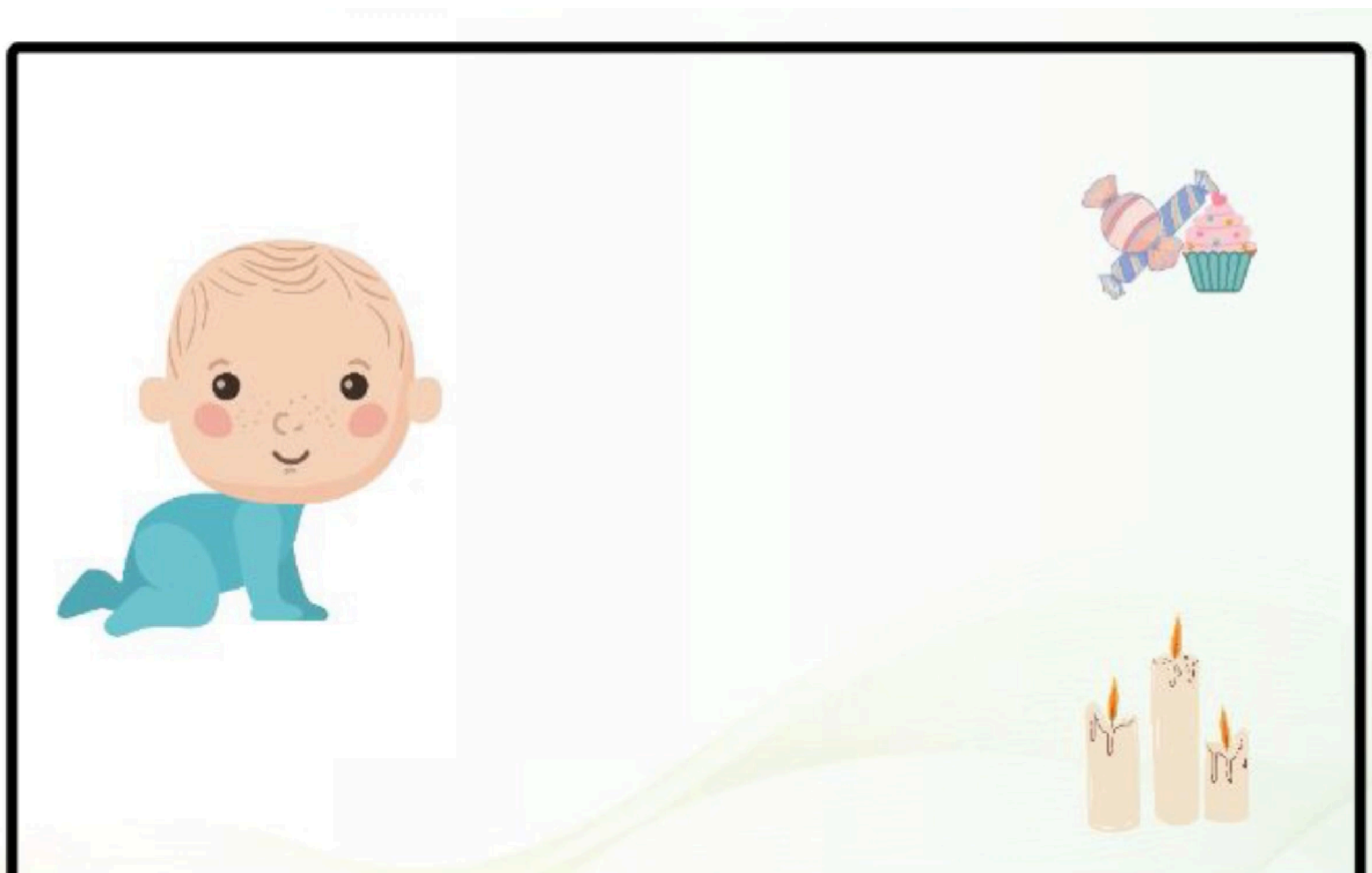
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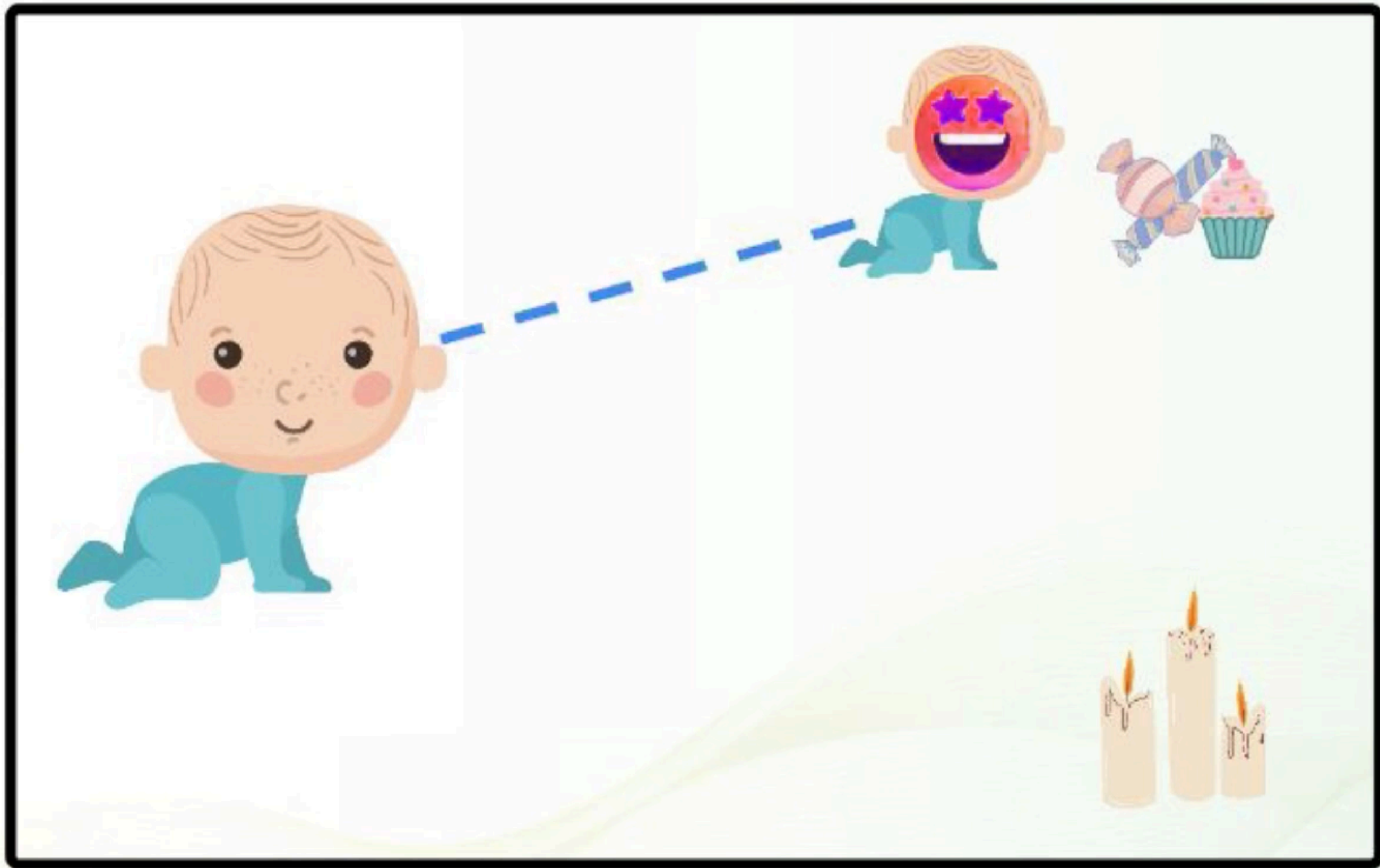
A

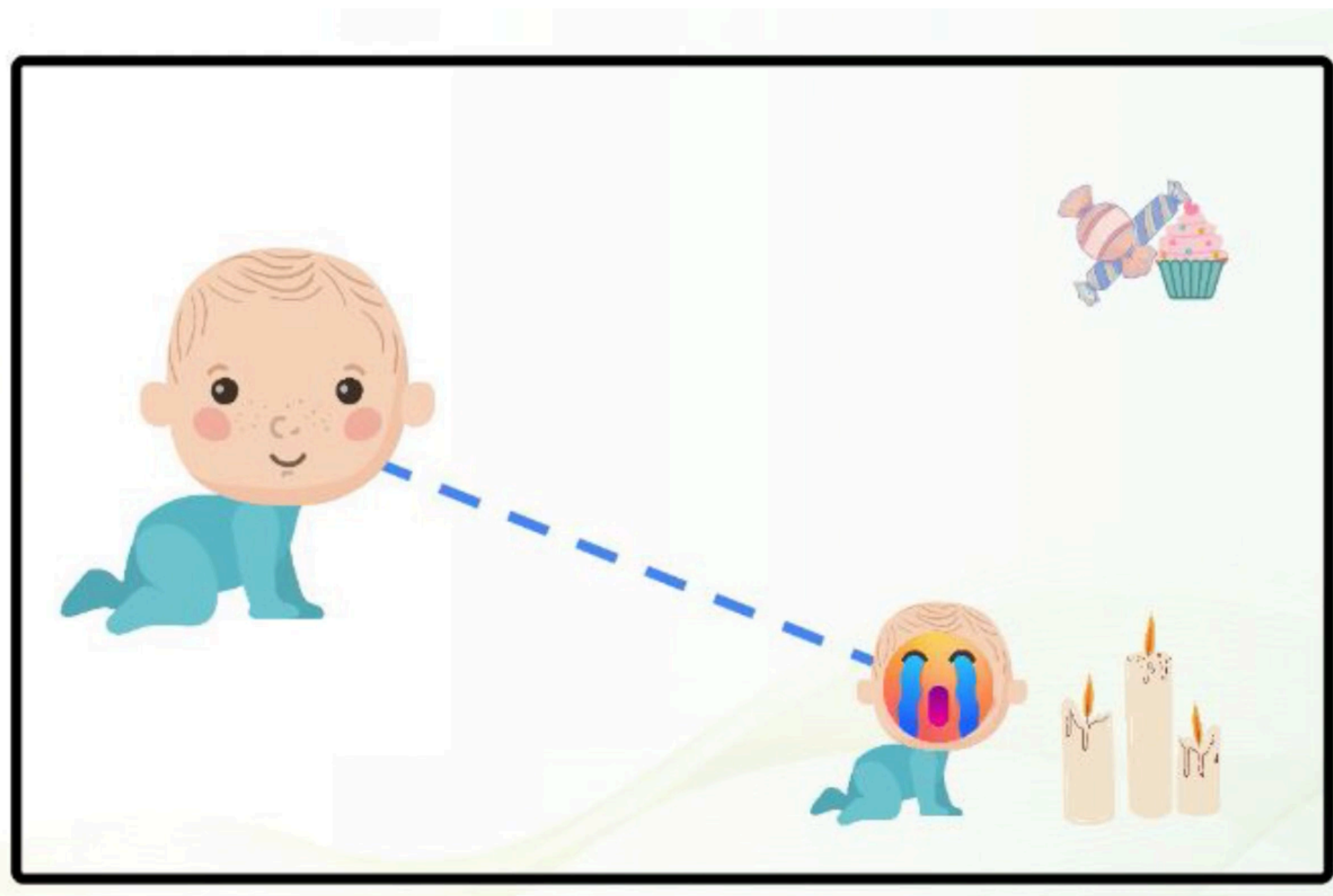
B

Path 1 = +60









Type 3: Reinforcement Learning

- Reinforcement Learning is a **feedback-based Machine learning technique** in which an **agent learns to behave in an environment by performing the actions and seeing the results of actions**.
- For each **good action**, the agent gets **positive feedback**, and for each **bad action**, the agent gets **negative feedback or penalty**.
- There is **no labelled data**, so the agent is bound to learn **by its experience** only.
- RL solves a **specific type of problem** where **decision making is sequential**, and the goal is long-term, such as game-playing, robotics, etc.

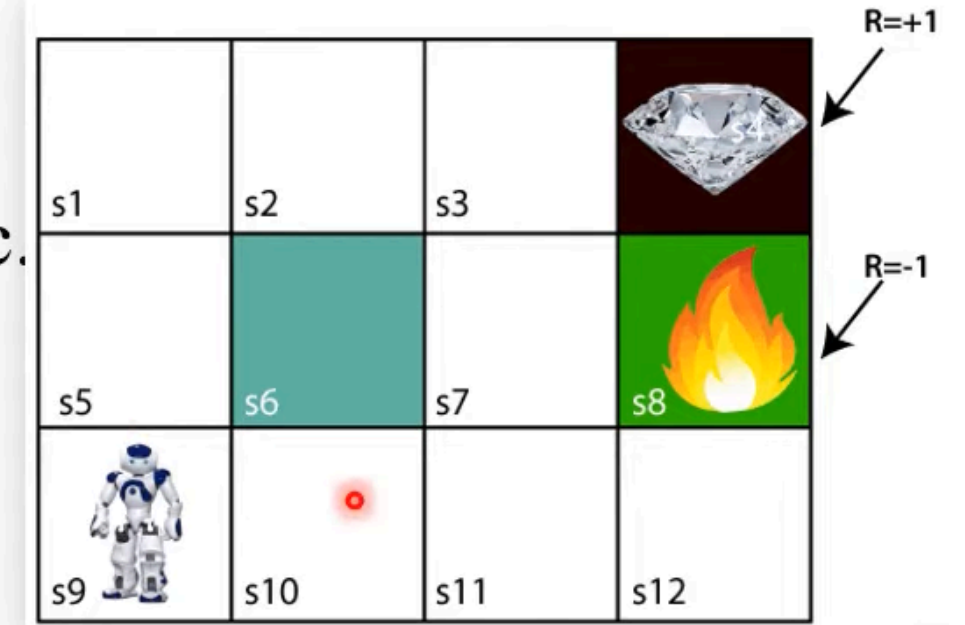


Working Example of Reinforcement Learning

- Consider two main things:

1. Environment: It can be anything such as a room, maze, ground etc.

2. Agent: An intelligent agent such as AI robot.



- In the image, the agent is at the very first S_9 block of the maze.
- The maze is consisting of an S_6 block, which is a **wall**, S_8 a **fire pit**, and S_4 a **diamond block**.
- The agent cannot cross the S_6 block, as it is a solid wall.
- If the agent reaches the S_4 block, then get the **+1 reward**; if it reaches the fire pit, then gets **-1 reward point**.
- It can take four actions: **move up, move down, move left, and move right**

Working of Reinforcement Learning

- The **agent** can take **any path** to reach to the **final point**, but he needs to make it in possible fewer steps.
- Suppose the agent considers the path **S9-S5-S1-S2-S3**, so he will get the **+1-reward** point.
- The agent will try to **remember the preceding steps** that it has taken to **reach the final step**.
- To memorize the steps, it **assigns 1 value** to each previous step.



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Applications of Reinforcement Learning

1. **Robotics:** RL is used in **Robot navigation, Robo-soccer, walking, juggling**, etc.
2. **Control:** RL can be used for **adaptive control** such as Factory processes, admission control in telecommunication, and Helicopter pilot is an example of reinforcement learning.
3. **Game Playing:** RL can be used in **Game playing** such as tic-tac-toe, chess, etc.
4. **Chemistry:** RL can be used for optimizing the chemical reactions.
5. **Business:** RL is now used for business strategy planning.
6. **Manufacturing:** In various automobile manufacturing companies, the robots use deep reinforcement learning to pick goods and put them in some containers.
7. **Finance Sector:** The RL is currently used in the finance sector for evaluating trading strategies.



Difference between Supervised, Unsupervised & Reinforcement Learning

Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labeled data	The machine is trained on unlabeled data without any guidance	An agent interacts with its environment by performing actions & learning from errors or rewards
Type of problems	Regression & classification	Association & clustering	Reward-based
Type of data	Labeled data	Unlabeled data	No predefined data
Training	External supervision	No supervision	No supervision
Approach	Maps the labeled inputs to the known outputs	Understands patterns & discovers the output	Follows the trial-and-error method



Supervised Learning



Unsupervised Learning



Reinforcement Learning

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Is Reinforcement Learning Task-Focused?

👉 Yes. Reinforcement Learning (RL) is mainly focused on completing a specific task by maximizing reward.

